BLEEH TECH

Boiler Temperature Control System

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What is Boiler Temperature Control System ?

Boiler Temperature Control System is a SMART TEMPERATURE DETECTOR that will not only control the temperature , but also adjust the speed of the fan according to the temperature which will help automating the process.

HARDWARE PIN MAPPING

COMPONENTS NAME	CONNECTION
ANALOG TEMPERATURE SENSOR	PC0 (ADC0)
COOLING FAN MOTOR	PD6 (OC0A)
LOCAL HAZARD ALARM	PD1 (TXD)

Working

The system is designed to automatically monitor and regulate the temperature of the boiler using ATMEGA328P. The system continuously read's the BOILERS TEMPERATUER and adjust fan's speed using PWM (PULSE WIDTH MODULATION) & triggers both local physical alarm and a wireless serial alarm IF TEMPERATURE EXCEEDS 80 degree Celsius. When the temperature in the range.

Subsystem Configuration (-_-)

Temperature sensing ADC module :

To interpret the analogue voltage coming from the temperature sensor the system utilizes the microcontrollers embedded 10 bit analogue to digital converter (ADC).

- The ADC provides a 10 bit digital representation of analogue signal.
- The system uses the ADC_Read() functions to trigger a convertion and return the digital value which the software then mathematically converts inro celsius temperature.

Fan Speed Control (PWM Module):

- The hardware timer blocks inside the ATMEGA329P facilitate the generation of PWM (Pulse Width Modulation) signals to control the fan's speed.
- The system operates in Fast PWM mode. By altering the value in the output compare register (OCR0A) the system adjusts the duty cycle of teh signal sent from pin PD6.

Wireless Hazard Alerts(USART Module) For wireless alerts , the system uses the Universal Synchronous and Asynchronous Receiver and Transmitter (USART) module to communicate serially, sending data out one bit at a time over the TX pin. The code configures several internal registers to achieve this:

Baud Rate: The transmission speed is set to 9600 bits per second. This is achieved by loading a calculated value (103 for a 16MHz clock) into the **UBRR0H** (high) and **UBRR0L** (low) registers.

- **Control and Frame Format:** The transmitter is enabled using the **UCSR0B** register. The system operates in Asynchronous mode (no external clock), and configures an 8-bit data frame format using the **UCSR0C** register.
- **Data Transmission:** When transmitting the warning message, the software first checks the **UCSR0A** register. Specifically, it looks at the **UDRE0** (USART Data Register Empty) bit to ensure the buffer is ready for a new byte. Once empty, an 8-bit character is loaded into the **UDR0** data register, which automatically pushes the data out to the wireless module

Software Logic & Program Flow 👽

Software Logic & Program Flow Upon powering up, the microcontroller initializes the ADC, PWM, and USART modules, and sets pin PD2 as an output. The software then enters an infinite while(1) loop, continuously performing the following routine:

1. **Read:** Samples the analog sensor and calculates the current temperature in Celsius.

2. **Actuate Fan:**
   - **< 35°C:** Fan is turned OFF (0% PWM).
   - **35°C – 49°C:** Fan runs at ~30% max speed.
   - **50°C – 70°C:** Fan runs at ~60% max speed.
   - **71°C – 100°C:** Fan runs at 100% max speed.
3. **Check Hazards:** Evaluates if the temperature has exceeded 80°C.
   - If **True**, it pulls pin PD2 HIGH to trigger the local alarm and utilizes the USART_SendString() function to transmit the warning message sequentially through the UDR0 buffer. A 1-second delay is introduced to prevent message spamming.
   - If **False**, pin PD2 is pulled LOW to silence any active local alarms.
4. **Wait:** Introduces a brief 100ms delay to stabilize the system before taking the next temperature reading.